TING-YUAN SUNG

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Summary

Expertise experience in developing complex software from requirement definition, feature and functionality design, code implementation, unit test, QA and configuration management.

- ➢ **Design effective data structure and algorithm with balance memory and run time performance.**
- ➢ **Proficient in C/C++, PERL programming, data structure and graph algorithm.**
- ➢ **Enthusiastically pursuing root cause of failures and implement solutions.**
- ➢ **Strong problem solving skill, solid knowledge in debugging software/hardware issue.**

PROFESSIONAL EXPERIENCE

Xilinx Inc., San Jose, CA 2016 - 2017

Digital Design Engineer

Building SoC RTL connectivity model, integration RTL components from multiple design teams, developing custom tools and methodologies to improve development efficiency and quality.

- Build and integration of HDIO RTL modules.
- Stub IP cores generation and integration for fast turnaround QA.
- Generating power ports for SoC setlist based on IP's UPF definition.
- Filelist generation for VCS compilation.

Intel Inc., Santa Clara, CA 2012 - 2016

Design Automation Engineer

SoC/IP pre-silicon design validation and flow automation from control register management (SystemRDL), RTL code generation, Testbench/OVM generation, VCS build and simulation, Lint/Spyglass/Synthesis check. Quality Gatekeeper role by managing code check in and publish daily simulation QA result.

- Convert the IP group register management legacy data to the industry standard SystemRDL language. On time delivery this project when it was months delay when I joined project by re-written 4000 lines of perl code and fortified the continuous failing QA test suites.
- Automate the process and enable the QA suites running by batch scheduling, monitor job status, collecting simulation result and compile result on the group web page.
- Help to reduce the overall 30% QA run time by merging Git repos which also reduce the number of QA testcases. Merge not only Git repos but also the Git database for future reference.

Synopsys Inc., Sunnyvale, CA 2005 – 2012

Staff R&D Engineer

- Maintain Partitioning and Isomorphic Matching program in HSIM which generate input data for Hierarchical Solver.
- Custom Model Interface (CMI) in HSIM engine – Work with HSPICE model group in defining and implementing interface to HSIM, handling CMI isomorphic matching, and checking transient simulation accuracy on internal accuracy criterion.
- Stuck-at fault simulation with .force/.release in SPICE netlist – Enhancement of HSIM to resolve state splitting and state merging issues during transient simulation.
- Macro-model binning for statistical parameters "vth0" "u0" – Work with Parser and Model group in defining and handling such instance parameters for macro-model in isomorphic matching. Checking the model accuracy on internal accuracy criterion.
- Full chip power rail simulation (PWRA) – Target on reduce overall run time for full chip power simulation. Debug the circuit and identified where the bottleneck for transient simulation is.
- MRAM device integration – Study the user defined characteristic of MRAM device and implement a scheme to handle such kind of current-controll-resistor. Implement code for device

stamping and resolve the related numerical issue during the DC and transient simulation.
- Lead the quality improvement project for HSIM – Coordinate among several groups – parser, model, CircuitCheck core simulation – to improve overall turnaround time for code check-in QA test based on Puricov database.

Nassda Inc., Santa Clara, CA 2002 – 2005

Sr. Product Specialist

- HSIM product specialist – Circuit debugging for various types of mixed analog and digital circuit including
  SRAM, DRAM, Charge Pump, A2D, D2A when simulation failed in HSIM engine
- Continuous support various model accuracy used in HSIM engine.
- Primary focal point support for Asia, Japan, and Korea customers.

Synplicity Inc., Sunnyvale, CA 1999 – 2001

Sr. Algorithm Developer

- Support placement based timing engine used in Altera/Xilinx mapper.
- Extract timing files for the Altera and Xilinx FPGA devices based on routing experiment result.
- Design floorplan scheme to include all cirtical path instances and redo the synthesis to improve the critical path timing.

Additional Information

National Semiconductor, Sunnyvale, CA

Design Automation Engineer

- Mixed analog and digital design. Study and evaluate tools to automate analog layout.
- Evaluation of RC extraction software and DRC/ERC tools from different vendors.
- VerilogAssem – Design Verilog parser to read in Verilog netlist, generate a new P&R acceptable Verilog netlist. This program can also insert power/ground net to do power analysis.

Silvar Lisco Inc., Sunnyvale, CA

Software Engineer

- Global Router. Works include generating topology graph according floorplan and placement. Each edge of this graph is associated with an edge cost which based on its length and the channel density. A graph based maze router was used to find the minimum spanning tree for each net. Afterward, there are several minimization steps to optimize density of the critical channels in order to reduce the overall die size.
- Pre-routing delay estimation. Based on routing data from global router, generate its equivalent RC circuit and calculate the wire delay.

Education

M.S. in Electrical and Computer Engineering

University of Colorado, Boulder, CO

B.S. in Electronic Engineering

Chung-Yuan Christian University, Chung-Li, Taiwan